

WEIGHT FORENSICS: DECODING TRAINING DATA DISTRIBUTION PROPERTIES DIRECTLY FROM NEURAL NETWORK WEIGHTS

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ABSTRACT

The proliferation of publicly deployed neural networks raises urgent questions about what these models encode about their training data. While membership inference and model inversion attacks probe individual data points, a more practically important and underexplored question is: can we decode *distributional* properties of training data—such as class imbalance ratios and label noise rates—directly from model weights, without any access to training data or model queries? We propose **Weight Forensics**, a framework that treats this as a supervised weight-space regression problem. We construct a model zoo of 300 small CNNs trained on controlled variants of CIFAR-10, CIFAR-100, Fashion-MNIST, and SVHN with systematically varied distributional parameters, and train MLP meta-models to predict these properties from per-layer weight statistics augmented with SVD spectral features. Across 8 dataset–property combinations evaluated via 5-fold cross-validation, we achieve mean test R^2 of 0.77–0.88. However, we uncover a critical pitfall: mild class imbalance (ratio ≤ 3) is essentially undetectable ($R^2 \approx 0$), cross-dataset transfer degrades sharply when domain statistics differ, and the forensic signal is entangled with dataset-specific feature statistics. These findings expose both the promise and the fundamental limits of weight-based data auditing.

1 INTRODUCTION

As machine learning models are increasingly deployed in high-stakes settings, regulatory frameworks demand accountability for training data properties—including demographic representation, class balance, and label quality. Yet once a model is deployed, the training data is rarely available for inspection. Auditors must rely on whatever artifacts remain: typically, only the model weights themselves.

Existing privacy attacks probe *individual* data points: membership inference (??) asks whether a specific sample was used in training, while model inversion (?) attempts to reconstruct representative inputs. Both require query access to the model. A complementary and arguably more practically relevant question has received far less attention: can we decode *distributional* properties of training data—class imbalance ratios, label noise rates—purely from the weights of a deployed model, with no data access and no model queries?

We introduce **Weight Forensics**, a framework for this task. Our approach treats distributional property prediction as a supervised regression problem over a *model zoo*: a large collection of networks trained under systematically varied distributional conditions. A meta-model is trained to predict ground-truth labels from weight-space features. Our findings are encouraging but sobering. Weight statistics *do* encode meaningful forensic signals: across four datasets and two distributional properties, we achieve mean 5-fold cross-validated R^2 of 0.77–0.88. However, we identify three critical limitations: (1) a sharp *detectability threshold*—mild imbalance (ratio ≤ 3) is completely undetectable; (2) *cross-dataset transfer* degrades substantially when domain statistics differ; and (3) the forensic signal is *entangled with dataset-specific features* in ways that limit generalization. These

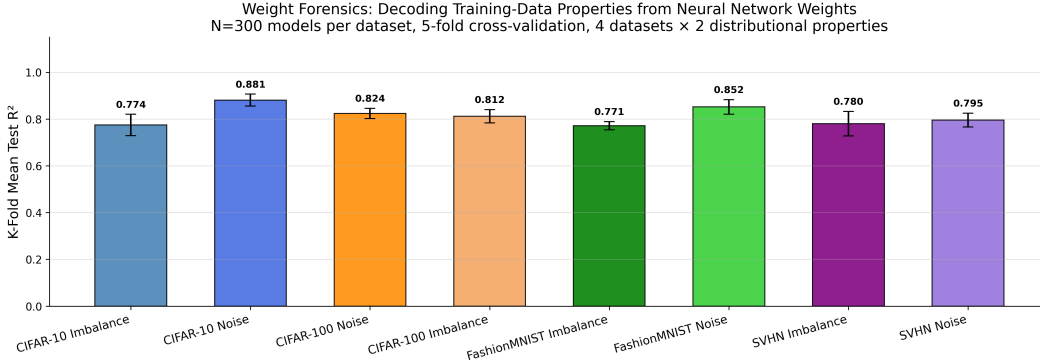


Figure 1: 5-fold CV mean test R^2 ($N = 300$ models per dataset) for decoding class imbalance ratio and label noise rate from weights across 8 dataset–property combinations. All tasks achieve $R^2 \geq 0.771$. Noise rate is consistently more decodable than imbalance ratio. Error bars show ± 1 std across folds.

negative and inconclusive results are precisely what the community needs before deploying weight-based auditing tools.

2 RELATED WORK

Privacy attacks on trained models. Membership inference attacks (??) determine whether a specific sample was in the training set, requiring query access. Model inversion (?) reconstructs representative inputs from confidence scores. Dataset inference (?) verifies dataset ^{Anonymous}ship but also requires query access. All target individual samples rather than distributional properties, and none operate purely from weights.

Weight-space learning. Hyper-representations (?) learn self-supervised embeddings of neural network weights for predicting model characteristics. Neural Functional Networks (?) and graph neural networks (?) define permutation-equivariant architectures for processing weight tensors. Model zoos (???) have been used for weight-space learning, interpolation, and population analysis. None of these works target the forensic decoding of training data distributional properties.

3 METHOD

Problem formulation. Given only the weights θ of a trained model, predict a scalar distributional property y of its training data (e.g., class imbalance ratio or label noise rate), treated as a supervised regression problem using a model zoo.

Model zoo construction. For each dataset, we train $N = 300$ small CNNs (TinyCNN: 3 conv layers with batch norm, adaptive pooling, dropout) under randomly sampled distributional conditions. For class imbalance, minority classes are subsampled at rate $1/r$ where $r \sim \text{Uniform}(1, 15)$. For label noise, a fraction $\eta \sim \text{Uniform}(0, 0.35)$ of training labels are randomly corrupted.

Weight feature extraction. For each parameter tensor, we compute 14 per-layer statistics (mean, std, percentiles, sparsity, skewness, kurtosis) and 5 SVD-based spectral features for 2D weight matrices (largest/smallest singular value, condition number, energy fraction in top-4 singular values, effective rank). Features are concatenated across all layers and z-score normalized.

Meta-model. A 4-layer MLP (hidden sizes 256, 128, 64 with LayerNorm, ReLU, dropout) predicts the target property from the weight feature vector using MSE loss, Adam optimizer, and cosine annealing, evaluated via 5-fold cross-validation.

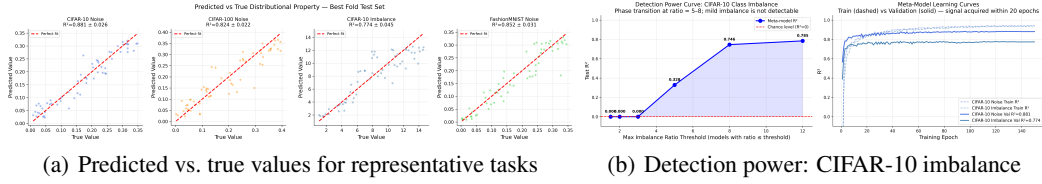


Figure 2: (a) Predicted vs. true distributional property values on the best fold test set. Points cluster near the identity line, though saturation at extreme imbalance values is visible. (b) A sharp phase transition in detectability: mild imbalance (ratio ≤ 3) is completely undetectable ($R^2 = 0$), moderate imbalance (ratio ≥ 8) is reliably decodable ($R^2 = 0.746$), and the intermediate regime (ratio ≈ 5) shows only weak decodability ($R^2 \approx 0.325$).

4 EXPERIMENTS

Main results. Figure 1 shows the primary result: across all 8 dataset–property combinations, our meta-model achieves $R^2 \in [0.771, 0.881]$. Label noise rate is consistently more decodable than class imbalance ratio on every dataset (CIFAR-10: 0.881 vs. 0.774; CIFAR-100: 0.824 vs. 0.812; FashionMNIST: 0.852 vs. 0.771; SVHN: 0.795 vs. 0.780). Label noise likely leaves a more broadly distributed footprint—corrupting gradient directions across all classes—whereas imbalance effects may be partially absorbed by last-layer biases and calibration.

Detection power and fundamental limits. Figure 2(b) reveals a critical pitfall for practical deployment. Restricting the CIFAR-10 imbalance zoo to models with ratio $\leq \tau$ and measuring R^2 as a function of τ exposes a sharp phase transition: $R^2 = 0$ for $\tau \leq 3$, rising to 0.325 at $\tau = 5$, and 0.746 at $\tau = 8$. Mild class imbalance—most common in real-world datasets and of greatest regulatory concern—is essentially undetectable from weights alone. The scatter plots in Figure 2(a) further reveal systematic underestimation at extreme imbalance ratios, suggesting saturation in the meta-model’s dynamic range.

Cross-dataset transfer. Figure 3 shows the cross-dataset transfer R^2 matrix. In-distribution performance (diagonal) is near-perfect ($R^2 \approx 0.98$), but off-diagonal transfer is highly asymmetric. Natural-image datasets transfer moderately to each other (CIFAR-10→SVHN: 0.793; SVHN→CIFAR-100: 0.720), but FashionMNIST transfers poorly both as a source and as a target (values of 0.10–0.42). Weight patterns encoding imbalance in grayscale clothing images are qualitatively different from those in natural RGB images, severely limiting the generalizability of a single forensic model.

Ablation studies. Adding SVD spectral features provides consistent gains (mean $+0.064 R^2$), with the largest improvement on FashionMNIST imbalance ($+0.155$; Appendix Figure 4). A 4-layer MLP outperforms a 1-layer MLP by $+0.10$ to $+0.21 R^2$ across tasks (Appendix Figure 5), confirming that non-linear feature interactions matter. Forensic decodability improves with CNN training duration, saturating around 12 epochs for noise tasks while imbalance tasks continue improving to 24 epochs (Appendix Figure 6). Zoo size ablation shows performance saturates around $N = 150$ – 200 models (Appendix Figure 7), justifying our choice of $N = 300$. Feature normalization is critical: no normalization collapses imbalance decoding to $R^2 \approx 0.23$ – 0.40 , while Z-Score and Min-Max normalization both perform well (Appendix Figure 8). Gradient and temporal features provide inconsistent benefits, with temporal cues modestly helping SVHN imbalance ($+0.026$) but consistently hurting noise rate decoding (Appendix Figures 9–10).

5 CONCLUSION

We presented Weight Forensics, a framework for decoding distributional training data properties directly from neural network weights. Our experiments across four datasets demonstrate that class imbalance ratios and label noise rates are reliably encoded in weights, with mean R^2 of 0.77–0.88 under 5-fold cross-validation. However, three important pitfalls emerge: mild imbalance ($\leq 3\times$) is completely undetectable; forensic models do not generalize across datasets with different domain statistics; and label noise is substantially more decodable than class imbalance. Future work

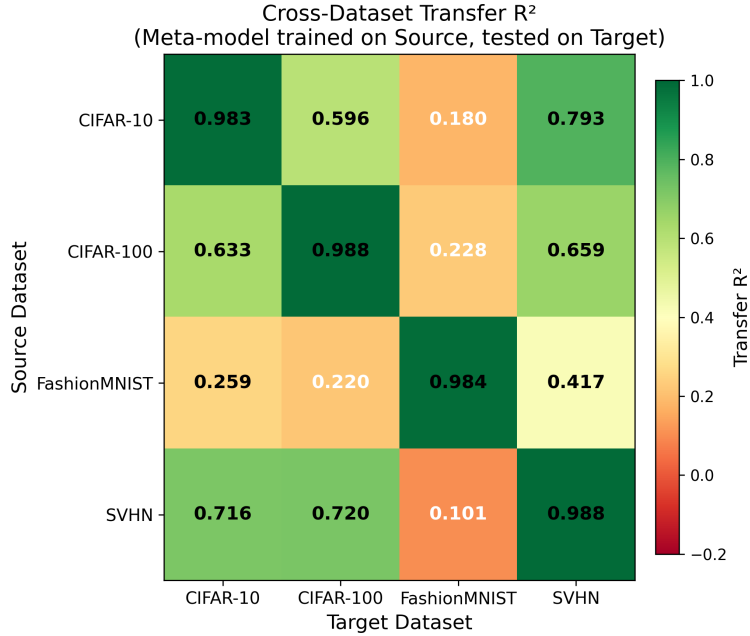


Figure 3: Cross-dataset transfer R^2 matrix for imbalance decoding. Diagonal (in-distribution) entries are near-perfect ($R^2 \approx 0.98$), but off-diagonal transfer is highly asymmetric: natural-image datasets transfer moderately to each other (e.g., CIFAR-10→SVHN: 0.793), while FashionMNIST transfers poorly both as source and target (values of 0.10–0.42), indicating that the forensic signal is entangled with domain-specific feature statistics.

should explore permutation-equivariant weight-space architectures (??) for better cross-architecture generalization, multi-source training for domain-robust forensic models, and calibrated uncertainty estimates for auditors. We hope these findings help the community set realistic expectations for weight-based data auditing in regulatory contexts.

REFERENCES

SUPPLEMENTARY MATERIAL

A ABLATION STUDIES

B BASELINE RESULTS

C ADDITIONAL ABLATIONS

D MUTUAL INFORMATION FEATURE SELECTION

E EXPERIMENTAL DETAILS

Model zoo. For each dataset, we train $N = 300$ TinyCNN models. The TinyCNN architecture consists of three convolutional blocks (Conv2d, BatchNorm2d, ReLU, MaxPool or AdaptiveAvgPool), followed by a two-layer MLP classifier with dropout. For CIFAR-10 and SVHN (32×32 RGB), we use 16/32/64 channels; for Fashion-MNIST (28×28 grayscale), we use a smaller variant with 16/32 channels; for CIFAR-100, we use 32/64/128 channels. All models are trained for 6 epochs with Adam (lr=1e-3, weight decay=1e-4) and cosine annealing.

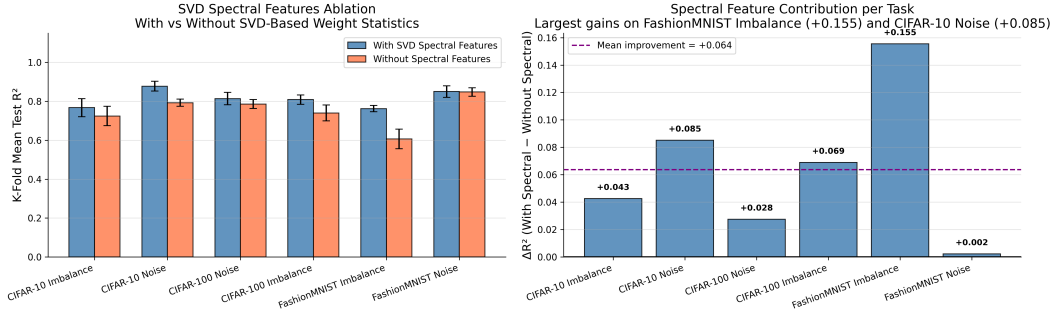


Figure 4: SVD spectral features ablation. Left: K-fold mean test R^2 with and without SVD-based spectral features across 6 dataset–property combinations. Right: ΔR^2 per task (mean gain +0.064, dashed line). Spectral features consistently improve performance, with the largest gain on FashionMNIST imbalance (+0.155) and negligible gain for FashionMNIST noise (+0.002), indicating spectral features are most useful for imbalance decoding.

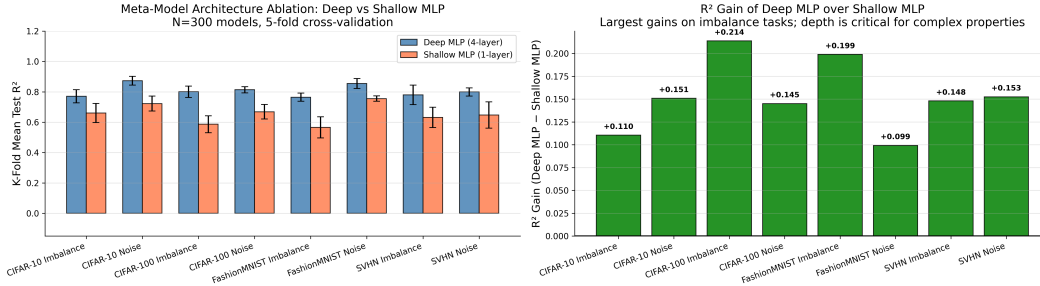


Figure 5: Meta-model architecture ablation: deep MLP (4-layer) vs. shallow MLP (1-layer). Left: absolute R^2 values across all 8 dataset–property combinations. Right: R^2 gain of deep over shallow MLP per task. Depth provides substantial gains (+0.10 to +0.21), with the largest improvements on CIFAR-100 Imbalance (+0.214) and FashionMNIST Imbalance (+0.199), confirming that non-linear feature interactions are critical.

Distributional conditions. For class imbalance, we designate 5 majority and 5 minority classes, sampling majority classes at $n_{\text{maj}} = 300\text{--}500$ examples and minority classes at $n_{\text{min}} = n_{\text{maj}}/r$ where $r \sim \text{Uniform}(1, 15)$. For label noise, we randomly relabel a fraction $\eta \sim \text{Uniform}(0, 0.35)$ of training examples with uniform random labels. For CIFAR-100, we use $n_{\text{per class}} = 35$ and imbalance ratios $r \sim \text{Uniform}(1, 10)$.

Meta-model training. We use 5-fold cross-validation with the same random seed (42) across all experiments. Within each fold, we train for 150 epochs with Adam (lr=1e-3, weight decay=5e-4) and cosine annealing. Targets are normalized to $[0, 1]$ before training and denormalized for evaluation. We report mean and standard deviation of test R^2 across folds.

Datasets. We use CIFAR-10 and CIFAR-100 (?), Fashion-MNIST (?), and SVHN (Street View House Numbers). All images are normalized with dataset-specific mean and standard deviation.

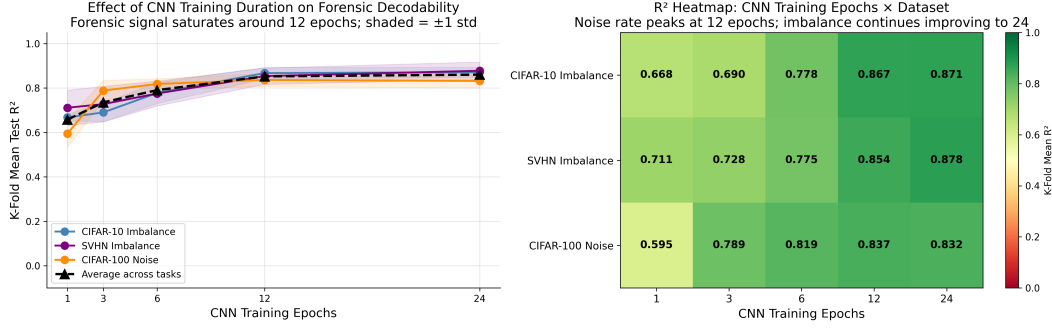


Figure 6: Effect of CNN training duration on forensic decodability for three representative tasks (CIFAR-10 Imbalance, SVHN Imbalance, CIFAR-100 Noise). Left: R^2 vs. training epochs (1, 3, 6, 12, 24) with ± 1 std shading. Right: heatmap of R^2 values. CIFAR-100 Noise peaks at 12 epochs (0.837) and slightly decreases at 24 (0.832), while imbalance tasks continue improving through 24 epochs, suggesting longer training may wash out noise-related weight signatures while reinforcing imbalance-related ones.

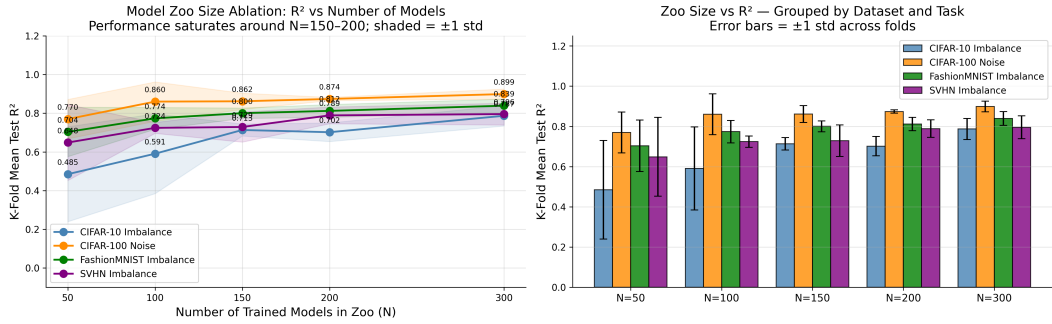


Figure 7: Model zoo size ablation for four representative tasks. Performance generally increases with N and begins to saturate around $N = 150-200$ models for most tasks, justifying our choice of $N = 300$. Both panels show the same data as a line chart (left, with ± 1 std bands) and grouped bar chart (right).

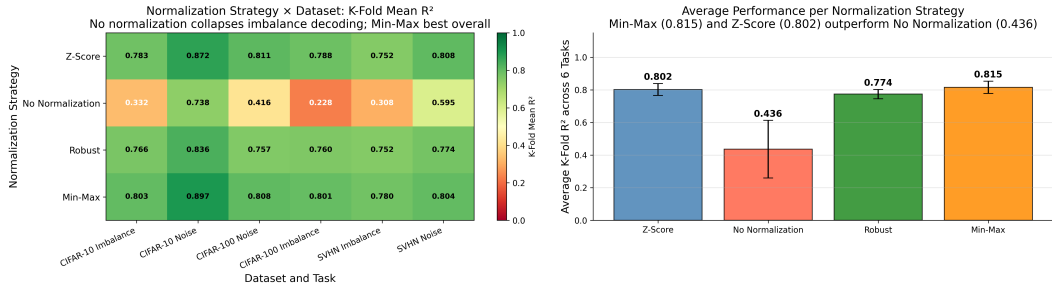


Figure 8: Feature normalization ablation across 6 dataset–property combinations. Left: heatmap of K-fold mean R^2 per normalization strategy. No normalization collapses imbalance decoding ($R^2 \approx 0.23-0.40$) while performing moderately on noise tasks. Right: average performance per strategy—Min-Max (0.815) and Z-Score (0.802) are both effective; we use Z-Score throughout for robustness to outliers.

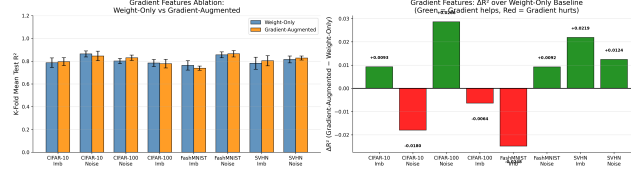


Figure 9: Gradient features ablation: ΔR^2 (gradient-augmented minus weight-only baseline) per dataset–property combination. Green bars indicate improvement, red bars indicate degradation. Adding gradient statistics provides inconsistent benefits (ΔR^2 from -0.025 to $+0.022$), suggesting gradient information is partially redundant with or noisy relative to weight statistics alone.

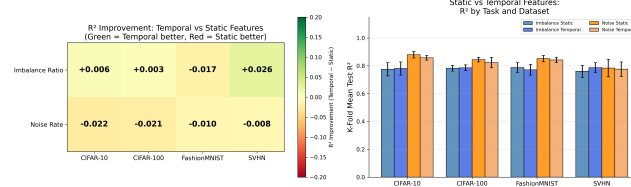


Figure 10: Temporal features ablation: R^2 improvement over static final weights when using weight snapshots at multiple training checkpoints. Temporal cues modestly help imbalance decoding on SVHN ($+0.026$) but consistently hurt noise rate decoding across all four datasets (-0.008 to -0.022), suggesting weight trajectories encode complementary information for imbalance but obscure noise-related signatures.

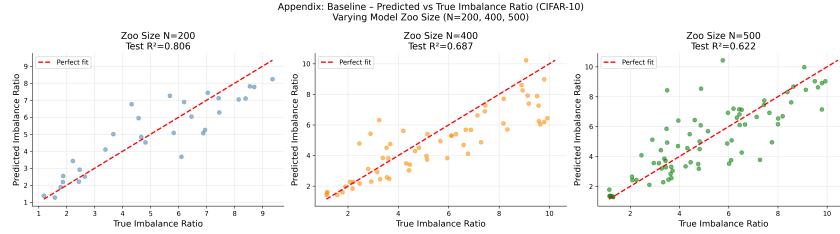


Figure 11: Baseline experiment on CIFAR-10 imbalance: predicted vs. true imbalance ratio for varying model zoo sizes ($N = 200, 400, 500$) using simple weight statistics without spectral features. Test R^2 counterintuitively decreases with zoo size ($0.806 \rightarrow 0.687 \rightarrow 0.622$), highlighting sensitivity to zoo construction—a key pitfall motivating stratified 5-fold cross-validation in all subsequent experiments.

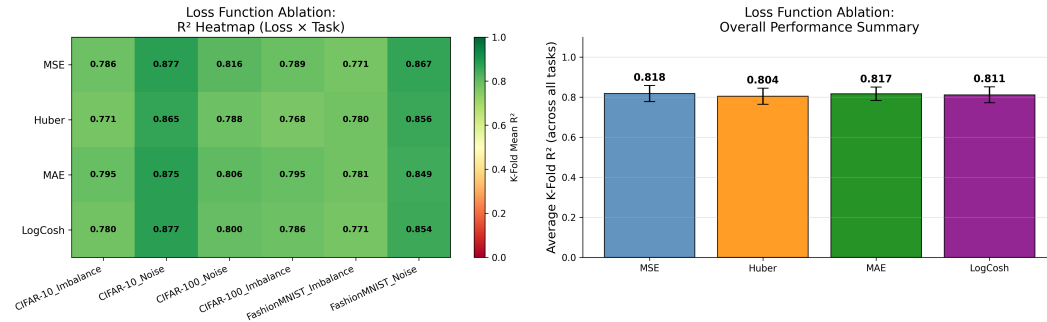


Figure 12: Loss function ablation: average K-fold R^2 across all tasks for MSE (0.818), Huber (0.804), MAE (0.817), and LogCosh (0.811). Differences between loss functions are small (range ≈ 0.014), confirming that the forensic signal is robust to the choice of regression objective. We use MSE throughout.

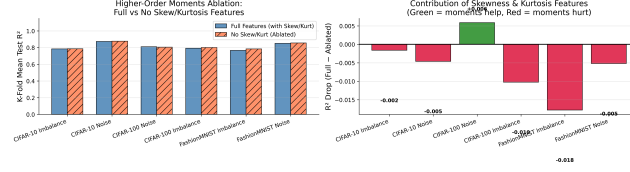


Figure 13: Higher-order moments ablation: R^2 drop when removing skewness and kurtosis features (positive = removal hurts, negative = removal helps). Skewness and kurtosis are largely redundant—their removal often slightly improves performance (e.g., FashionMNIST imbalance: -0.018), with only CIFAR-100 noise showing a small benefit from including them. This motivates future work on principled feature selection.

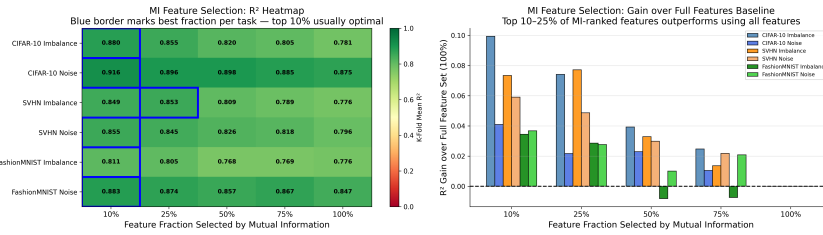


Figure 14: Mutual information (MI)-based feature selection: R^2 as a function of the fraction of features selected by MI ranking across six dataset–property combinations. Left heatmap: blue borders mark the best fraction per task—the top 10% of MI-ranked features is optimal for most tasks. Right bar chart: gain over the full feature set (100% baseline), showing that using only the top 10–25% of MI-ranked features matches or outperforms using all features, confirming that many weight statistics are redundant.